

Ungraded assignment solution

a.

$$B^{-1}b = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 20 \\ 30 \end{bmatrix} = \begin{bmatrix} -10 \\ 30 \end{bmatrix}, c_B B^{-1}b = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} -10 \\ 30 \end{bmatrix} = 60$$

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	1	1	0	2	60
0	0	-1	5	1	-1	-10
0	1	4	-1	0	1	30

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	0	6	1	1	50
0	0	1	-5	-1	1	10
0	1	0	19	4	-3	-10

z	x_1	x_2	x_3	x_4	x_5	RHS
1	1/3	0	37/3	7/3	0	140/3
0	1/3	1	4/3	1/3	0	20/3
0	-1/3	0	-19/3	-4/3	1	10/3

Optimal solution is $x_1^* = 0, x_2^* = 20/3, x_3^* = 0, z^* = 140/3$

b.

$$c_B B^{-1}A_3 - c_3 = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} - (-2) = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ -2 \end{bmatrix} + 2 = -2, \text{ not optimal (suboptimal).}$$

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	1	-2	0	2	20
0	0	-1	5	1	-1	20
0	1	4	-2	0	1	10

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	3/5	0	2/5	8/5	28
0	0	-1/5	1	1/5	-1/5	4
0	1	18/5	-0	2/5	3/5	18

Optimal solution is $x_1^* = 18, x_2^* = 0, x_3^* = 0, z^* = 28$

c.

$$c_B B^{-1} A_1 - c_1 = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} - 4 = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} - 4 = 0$$

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	1	1	0	2	20
0	1	-1	5	1	-1	20
0	2	4	-1	0	1	10

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	1	1	0	2	20
0	0	-3	11/2	1	-3/2	15
0	1	2	-1/2	0	1/2	5

Optimal solution is $x_1^* = 5, x_2^* = 0, x_3^* = 0, z^* = 20$

e.

z	x_1	x_2	x_3	x_4	x_5	RHS
1	-1	-5	2	0	0	0
0	0	-1	5	1	-1	20
0	1	4	-1	0	1	10

z	x_1	x_2	x_3	x_4	x_5	RHS
1	0	-1	1	0	1	10
0	0	-1	5	1	-1	20
0	1	4	-1	0	1	10

z	x_1	x_2	x_3	x_4	x_5	RHS
1	1/4	0	3/4	0	5/4	25/2
0	1/4	0	19/4	1	-3/4	45/2
0	1/4	1	-1/4	0	1/4	5/2

Optimal solution is $x_1^* = 0, x_2^* = 5/2, x_3^* = 0, z^* = 25/2$

f.

z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	0	1	1	0	2	0	20
0	0	-1	5	1	-1	0	20
0	1	4	-1	0	1	0	10
0	3	2	3	0	0	1	25

z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	0	1	1	0	2	0	20
0	0	-1	5	1	-1	0	20
0	1	4	-1	0	1	0	10
0	0	-10	6	0	-3	1	-5

z	x_1	x_2	x_3	x_4	x_5	x_6	RHS
1	0	0	8/5	0	17/10	1/10	39/2
0	0	0	22/5	1	-7/10	-1/10	41/2
0	1	0	7/5	0	-1/5	2/5	8
0	0	-1	-3/5	0	3/10	-1/10	1/2

Optimal solution is $x_1^* = 8, x_2^* = 1/2, x_3^* = 0, z^* = 39/2$

g.

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, B^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$c_B B^{-1} N - c_N = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 4 & 0 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} -7 & -3 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 7 & -3 & 0 \end{bmatrix} = \begin{bmatrix} -3 & 7 & 2 \end{bmatrix}$$

A question similar to the lab assignment using duality

Consider the following problem.

$$\begin{array}{ll}\min & c_2 x_2 - x_1 - x_3 \\ \text{s.t.} & x_3 - x_1 - x_2 \leq 1 \\ & 2x_1 - x_2 + x_3 \leq 2 \\ & -2 \leq x_1 \leq 1 \\ & -2 \leq x_3 \leq -1\end{array}$$

- (a) Construct the dual problem.
- (b) Use duality theory and explain if $(x_1, x_2, x_3) = (-1, 2, -1)$ can be an optimal solution. Does your answer depend on c_2 ? If so, provide necessary conditions.
- (c) Use duality theory and explain if $(x_1, x_2, x_3) = (1/3, -10/3, -2)$ can be an optimal solution. Does your answer depend on c_2 ? If so, provide necessary conditions.

Solution

a.

$$\begin{aligned} \min \quad & -x_1 + c_2x_2 - x_3 \\ \text{s.t.} \quad & -x_1 - x_2 + x_3 \leq 1 \end{aligned}$$

$$\begin{aligned} 2x_1 - x_2 + x_3 &\leq 2 \\ x_1 &\geq -2 \\ x_1 &\leq 1 \\ x_3 &\geq -2 \\ x_3 &\leq -1 \\ x_1, x_2, x_3 &\text{urs} \end{aligned}$$

$$\begin{aligned} \max \quad & y_1 + 2y_2 - 2y_3 + y_4 - 2y_5 - y_6 \\ \text{s.t.} \quad & -y_1 + 2y_2 + y_3 + y_4 = -1 \\ & -y_1 - y_2 = c_2 \\ & y_1 + y_2 + y_5 + y_6 = -1 \\ & y_1, y_2, y_4, y_6 \leq 0, y_3, y_5 \geq 0 \end{aligned}$$

b.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
-1	2	-1	3	7	1	2	1	0

The solution is feasible but it is not a feasible corner point. For it to be a feasible corner points there must be 3 non-basic variables but there is only one non-basic variable. That's why this solution cannot be a optimal solution and doesn't depend on value of c_2 .

c.

x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
1/3	-10/3	-2	0	0	7/3	2/3	0	1

This solution is a feasible and a corner point. To check whether it is optimal or not we should calculate its corresponding dual variables.

y_1	y_2	y_3	y_4	y_5	y_6
$\frac{1-2c_2}{3}$	$\frac{-1-c_2}{3}$	0	0	-1+c ₂	0

This solution can be optimal if necessary conditions are provided for c_2 .

$$y_1 = \frac{1-2c_2}{3} \leq 0, c_2 \geq 1/2$$

$$y_2 = \frac{-1-c_2}{3} \leq 0, c_2 \geq -1$$

$$y_5 = -1 + c_2 \geq 0, c_2 \geq 1$$

While $c_2 \geq 1$, this solution is optimal solution.